Protected areas as last bulwark against deforestation in Madagascar

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Background

A majority of Madagascar's endemic plants and animals are encountered in forests (Fig. 1). Since the 19th century, deforestation poses a threat to terrestrial biodiversity. In response, Madagascar was one of the first countries establishing protected areas (PAs) since 1927 and has since been expanding the network to encompass over 110 terrestrial PAs. Since the 1950s, Madagascar has lost almost half of its forests.

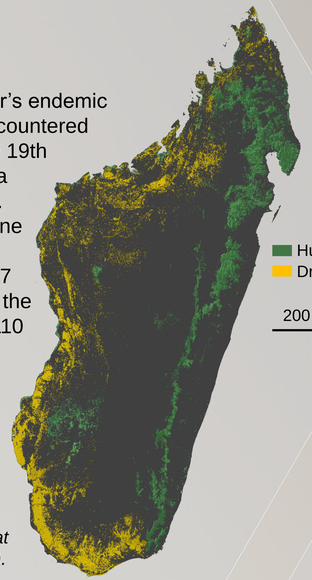
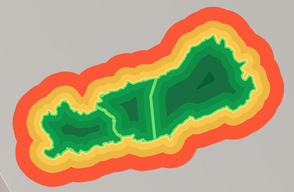


Fig. 1. Forest cover simplified from Moat and Smith (2007).

Methods

We estimated forest cover according to Vieilledent *et al*. (2018) within the PAs boundaries and outside for the years 1990, 2000, 2005, 2010, 2015 and 2017 (Fig. 2).



Inside PAs:	Outside PAs:
Buffer 1 (0–500m)	Buffer 1 (0–500m)
Buffer 2 (0.5–2.5km)	Buffer 2 (0.5–2.5km)
Buffer 3 (2.5–5km)	Buffer 3 (2.5–5km)
Buffer 4 (5–10km)	Buffer 4 (5–10km)
core areas	

Fig. 2. Calculation of the buffers inside and outside the PAs boundaries.

All PAs data and legal text have been obtained from Madagascar National Parks (cf. Rafanoharana *et al*.)

Results

The PAs covered an area of some 1.2 million hectares (M ha) in 1990. Since 2015, the terrestrial PAs network has increased by more than a factor 5 totaling 6.40 M ha for 123 PAs to date. Forest cover is increasing from the borders towards the core (Fig. 3a,b). The highest deforestation pressure exists within PAs in Buffer 1, amounting to some 20% loss between 1990 and 2017. The forest loss in Buffer 1 outside PAs (similar total area) was bigger than 30%. Forest cover within PAs is highest between Buffers 2–4, ranging from 60% to 66% in 1990, 66% to 75% in 2000, and 60 to 63% in 2015; 48% of the Buffer 1 inside PAs was covered with forest in 2015.

Forest cover (🌳🌳🌳) outside PAs is decreasing with increasing distance from the PAs' borders: e.g., in 2017 forest cover in Buffer 1 = 30%, in Buffer 2 = 22%, Buffer 3 = 17%, Buffer 4 = 12%.

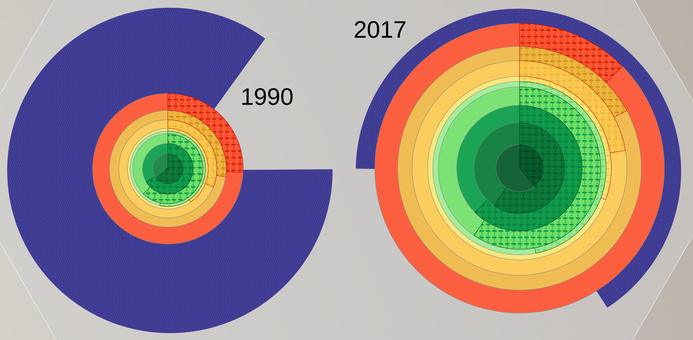


Fig. 3a. Areas of forest 🌳🌳 and other land cover according to distance to the borders of PAs in 1990 and 2017. (outer ring blue for land beyond outer buffers)

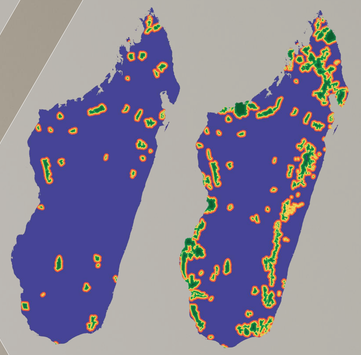


Fig. 3b. Expansion of the network of protected areas from 1990 to 2017 (distance from borders inside and outside of the PAs, as in the methods and Fig. 3a)

Conclusion

Given the high pressure on the forests in the first buffer inside the protected areas, the buffers just outside—which still hold some forests—are becoming increasingly important to attenuate the pressure within the protected areas.

References

Moat J, Smith P (Eds) 2007. Atlas of the vegetation of Madagascar. Royal Botanic Gardens, Richmond, England: Kew Publishing.

Rafanoharana SC, Andrianambinina FOD, Rasamuel HA, *et al*. Exemplifying deforestation processes in four protected areas in Madagascar. *Forests*. In press.

Vieilledent G, Grinand C, Rakotomalala FA, *et al*. 2018. Combining global tree cover loss data with historical national forest-cover maps to look at six decades of deforestation and forest fragmentation in Madagascar. *Biol. Conserv*. **222**: 189–97.

Acknowledgements

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A *Calumma parsonii*
B *Propithecus diadema*
C *Mantella aurantiaca*
D *Neodrepanis coruscans*

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